Homework 1

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QUESTION 1

(1)

$$P(\text{error} \mid \hat{y} = w_i) = P(\hat{y} = w_i) \cdot P \sum_{j \neq i} P(\hat{y} = w_j)$$

For a two class scenario (i.e, $i \in \{1, 2\}$):

$$\boxed{P(\text{error} \mid \hat{y} = w_i) = P(\hat{y} = w_i) \times (1 - P(y = w_i))}$$

1.1a

(2)

$$P(\text{error}) = f(p,q) = \sum_i P(\text{error} / \hat{y} = \omega_i)$$

$$= P(\text{error} / \hat{y} = \omega_1) + P(\text{error} / \hat{y} = \omega_2)$$

$$= P(\hat{y} = \omega_1)(1 - P(y = \omega_1)) + P(\hat{y} = \omega_2)(1 - P(y = \omega_2))$$

We have

$$P(\hat{y} = \omega_1) = q \qquad P(y = \omega_1) = p$$

$$P(\hat{y} = \omega_2) = 1 - q \qquad P(y = \omega_2) = 1 - p$$

$$\Rightarrow P(\text{error}) = q \cdot (1-p) + p(1-q)$$

$$= q - qp + p - pq$$

$$\boxed{P(\text{error}) = p + q - 2pq} \longrightarrow (1.2a)$$

(3)

Substituting $q = 0.5$ in (1.2a) we get:

$$P(\text{error}) = p + 0.5 - 2 \times 0.5 \times p$$

$$= \cancel{p} + 0.5 - \cancel{p}$$

$$\boxed{P(\text{error}) = 0.5} \longrightarrow (1.3a)$$

From (1.3a) it is evident that $P(\text{error})$ does not depend on p if $q = 0.5$.

(4)

Let us assume $q = 0.5$ and say $q' = q + \delta$ where $-0.5 \leq \delta \leq 0.5$. Using q' in (1.2a), we get:

$$P'(\text{error}) = p + q' - 2pq'$$

$$= p + q + \delta - 2p(q + \delta)$$

$$= p + q - 2pq + \delta(1 - 2p)$$

Let $k = (1 - 2p)$

$\Rightarrow$ $k \geq 0$ if $2p \leq 1$ or $p \leq 0.5$

$k \leq 0$ if $2p \geq 1$ or $p \geq 0.5$

$$\boxed{P'(\text{error}) = P(\text{error}) + \delta k} \longrightarrow (1.4a)$$

We can see that $P'(\text{error}) > P(\text{error})$ if

$\delta < 0$ and $p > 0.5$ OR

$\delta > 0$ and $p < 0.5$

We can conclude that for both $\delta > 0$ and $\delta < 0$ there exist some values of $p < 0$ and $p > 0$ respectively for which $P'(\text{error}) > P(\text{error})$

(5)

From (1.4a) we can deduce the worst case for $q < 0.5$ and $q > 0.5$ as follows:

If $q < 0.5 \Rightarrow \delta < 0$:

$$P'(\text{error}) = P(\text{error}) - \delta(1-2p)$$

The error is maximized when $p = 1$

$$\boxed{P'(\text{error})_{max} = P(\text{error}) + \delta} \longrightarrow (1.5a)$$

If $q > 0.5 \Rightarrow \delta > 0$:

$$P'(\text{error}) = P(\text{error}) + \delta(1-2p)$$

The error is maximized when $p = 0$

$$\boxed{P'(\text{error})_{max} = P(\text{error}) + \delta} \longrightarrow (1.5b)$$

(1.5a) and (1.5b) are identical and can be minimized by choosing $\delta = 0$ or $q = 0.5$.

If guessing Y without prior knowledge of p, the max error can be minimized by choosing available clause with equal probability, independent of p.

QUESTION 2

(1)

From (1.2 a) we have:

$$P(error) = p + q - 2pq$$

$$\boxed{f(p,q) = p + q(1-2p)} \longrightarrow (2.1a)$$

(2.1a) is a straight line with slope $k = (1-2p)$

If $p \leq 0.5$ k is positive (increasing)

$p \geq 0.5$ k is negative (decreasing)

Given that $p \in [1,0]$ and $q \in [1,0]$, the min values of error are:

If $k \geq 0$ smallest error is when $q = 0$

$k \leq 0$ smallest error is when $q = 1$

$$\boxed{q^*(p) = \begin{cases} 0 & \text{if } p \leq 0.5 \\ 1 & \text{if } p > 0.5 \end{cases}} \longrightarrow (2.1b)$$

NOTE: When $p = 0.5$, $P(error)$ is the same for both values of q as shown in the answer to question 1(5).

(2)

From (1.2a) we have:

$$f(p,q) = P(\text{error}) = p + q - 2pq$$

Using (2.1b), if $p < 0.5$, $q = 0$

$$P(\text{error}) = p + 0 - 0$$

$$\boxed{P(\text{error}) = p}$$

Using (2.1b), if $p > 0.5$, $q = 1$

$$P(\text{error}) = p + 1 - 2p$$

$$\boxed{P(\text{error}) = 1 - p}$$

Putting this together, we have:

$$\boxed{P(\text{error}) = f(p,q^*) = \begin{cases} p & \text{if } p > 0.5 \\ 1-p & \text{if } p < 0.5 \\ 0.5 & \text{if } p = 0.5 \end{cases}}$$

(2.2a)

(3)

The value of A from Q1.3 : $\boxed{A = 0.5}$

From (2.2a) assuming $p \neq 0$, we have:

$$f(p, q^*) = p \quad \text{if} \quad p < 0.5$$

$$\therefore \boxed{f(p, q^*) < 0.5 \quad \text{if} \quad p < 0.5} \longrightarrow (2.3a)$$

$$f(p, q^*) = 1 - p \quad \text{if} \quad p > 0.5$$

$$\text{If } p > 0.5 \quad 1 - p < 0.5$$

$$\therefore \boxed{f(p, q^*) < 0.5 \quad \text{if} \quad p > 0.5} \longrightarrow (2.3b)$$

We can see from (2.3a) and (2.3b) that P(error) is always lesser than $A = 0.5$ for all values of $p \neq 0.5$, and is equal to A for $p = 0.5$.

Knowing the prior probability p allows us to make better guesses with lower P(error) when compared to a blind 50-50 guess that we made when p was unknown. At worst, our solution is just as good as the best guess we can make when p is unknown.

Question 3

(1)

According to Bayes Decision Rule:

$$\varphi(P(\omega_1|x)) = \begin{cases} 0 & \text{if } P(\omega_1|x) < P(\omega_2|x) \\ 1 & \text{if } P(\omega_1|x) > P(\omega_2|x) \end{cases}$$

(3.1a)

We also know that:

$$P(\omega_1|x) + P(\omega_2|x) = 1.0$$

$$P(\omega_1|x) = 1 - P(\omega_2|x) \quad \text{OR}$$

$$P(\omega_2|x) = 1 - P(\omega_1|x)$$

This implies that:

If $P(\omega_1|x) < P(\omega_2|x)$ then $P(\omega_1|x) < 0.5$

If $P(\omega_1|x) > P(\omega_2|x)$ then $P(\omega_1|x) > 0.5$

(3.1b)

Using (3.1a) in (3.1b) we get

$$\varphi(P(\omega_1|x)) = \begin{cases} 0 & \text{if } P(\omega_1|x) < 0.5 \\ 1 & \text{if } P(\omega_1|x) > 0.5 \end{cases}$$

(3.1c)

(3.1c) is similar to (2.1b), only reformulated using $P(\omega_1|x)$ instead of p.

If we break down $P(\omega_i|x)$, we get

$$P(\omega_i|x) = \frac{P(x|\omega_i) \cdot P(\omega_i)}{P(x)}$$

Given that the denominator is identical for $i \in 1, 2$, we see that our decision can be obtained purely by comparing numerators.

The numerator is also the product of the likelihood ($P(\text{evidence}|\text{class})$) and the prior probability ($P(\text{class})$)

This shows that the Bayes Decision Rule takes into account not just the prior probability, but also the evidence, which is more optimal than just considering priors.

The answer to Q3.2 will show that this approach minimizes error.

(2)

Using $P(w_i|x)$ in (2.2a) we get

$$P(\text{error}|x) = \begin{cases} P(w_1|x) & \text{if } P(w_2|x) \geq 0.5 \\ P(w_2|x) & \text{if } P(w_1|x) > 0.5 \\ 0.5 & \text{if } P(w_1|x) = P(w_2|x) = 0.5 \end{cases}$$

It follows that for values of x where the error is $P(w_1|x)$:

$$P(w_1|x) < 0.5 \quad \text{or} \quad P(w_1|x) < P(w_2|x) \qquad (3.2a)$$

Similarly, for values of x where the error is $P(w_2|x)$:

$$P(w_2|x) < 0.5 \quad \text{or} \quad P(w_2|x) < P(w_1|x) \qquad (3.2b)$$

We see from (3.2a) and (3.2b) that for any given x, $P(\text{error}|x)$ is the smaller of $P(w_1|x)$ and $P(w_2|x)$.

$$\therefore \quad P(\text{error}|x) = \min\left(P(w_1|x), P(w_2|x)\right) \qquad (3.2c)$$

(3)

We can re-state (2.2a) as:

$$P(\text{error}) = \min\left(P(\omega_1), P(\omega_2)\right)$$

Let ω_{min} be the class with the lowest prior

$$\Rightarrow \boxed{P(\text{error}) = P(\omega_{min}) = \int_{-\infty}^{\infty} P(\omega_{min})\, dx} \quad \text{(3.3a)}$$

When using the posterior, we have:

$$P(\text{error}) = \int_{-\infty}^{\infty} P(\text{error}/x)\, p(x)\, dx.$$

From (3.2c)

$$P(\text{error}/x) = \min\left(P(\omega_1/x), P(\omega_2/x)\right)$$

For each x, we choose $\omega_i = \omega'_{min}$

$$\Rightarrow P(\text{error}) = \int_{-\infty}^{\infty} P(\omega'_{min}/x) * p(x)\, dx$$

$$= \int_{-\infty}^{\infty} \frac{P(x/\omega'_{min}) * P(\omega'_{min})}{p(x)} * p(x)\, dx$$

$$= \int_{-\infty}^{\infty} P(\omega'_{min}/x)\, dx$$

$$\boxed{P(\text{error}) = \int_{-\infty}^{\infty} P(\omega'_{min}/x)\, dx} \quad \text{(3.3b)}$$

We can see from (3.3b) that we choose the smallest possible error for each x instead of integrating using the smallest prior for all x.

At worst, this should make the error as large as (3.3a) if the evidence does not provide additional information i.e $P(x|w_1) = P(x|w_2)$. However, if the evidence provides some discriminating information, there will be at least some values of x for which $P(w'_{min}) < P(w_{min})$.

This implies that

$$\int_{-\infty}^{\infty} P(\text{error}|x)\, p(x)\, dx \leq P(w_{min}).$$

QUESTION 4

(1)

Intuitively $\lambda_{ab} > \lambda_{bb}$. This is because the cost of making an incorrect decision must be greater than that for making the right decision.

If this were not true, the classification system (comprising the classifier and the resultant action-taking system) would be incentivized to make incorrect decisions.

(2)

Intuitively λ_{PN} must be much larger than λ_{NP}.

This is because classifying a poisonous mushroom as edible can lead to serious health consequences to the eater of the mushroom. The cost of this action (λ_{PN}) should therefore be much higher.

Conversely, classifying an edible mushroom as poisonous merely deprives the would-be eater of the mushroom an opportunity to eat it, but has no potential for adverse health consequences. It thus follows that the corresponding cost λ_{NP} should be lower.

(3) Definitions:

ω_P → Class of non-poisonous (edible) mushrooms

ω_N → Class of poisonous (inedible) mushrooms

λ_{PN} → Cost of predicting "non poisonous" when the mushroom is actually poisonous

λ_{PP} → Cost of predicting "non-poisonous" when the mushroom is actually non-poisonous

TPR → The probability of predicting "non-poisonous" when the mushroom is also non-poisonous.

FPR → The probability of predicting "non-poisonous" when the mushroom is poisonous.

TNR → The probability of predicting "poisonous" when the mushroom is also poisonous (= 1 - FPR)

FNR → The probability of predicting "poisonous" when the mushroom is non-poisonous (= 1 - TPR)

The expected risk is summing over all cases, the following:

(cost of predicting ω_i for case ω_j) ×
(probability of ω_j) ×
(rate at which the classifier predicts ω_j)

$\therefore$ Risk $= \lambda_{PP} \times P(\omega_P) \times$ (Rate of correctly predicting ω_P) +
$\lambda_{PN} \times P(\omega_N) \times$ (Rate of incorrectly predicting ω_P) +
$\lambda_{NN} \times P(\omega_N) \times$ (Rate of correctly predicting ω_N) +
$\lambda_{NP} \times P(\omega_P) \times$ (Rate of incorrectly predicting ω_N)

$\Rightarrow$ Risk $= \lambda_{PP} \cdot P(\omega_P) \cdot TPR + \lambda_{PN} \cdot P(\omega_N) \cdot FPR +$
$\lambda_{NN} \cdot P(\omega_N) \cdot TNR + \lambda_{NP} \cdot P(\omega_P) \cdot FNR$

$= \lambda_{PP} \times P(\omega_P) \times TPR + \lambda_{PN} \times P(\omega_N) \times FPR$
$\lambda_{NN} \times P(\omega_N)\,(1-FPR) + \lambda_{NP} \times P(\omega_P) \cdot (1-TPR)$

$= \lambda_{PP} \cdot P(\omega_P) \cdot TPR - \lambda_{NP} \times P(\omega_P) \cdot TPR +$
$\lambda_{PN} \cdot P(\omega_N) \cdot FPR - \lambda_{NN} \times P(\omega_N) \cdot FPR +$
$\lambda_{NN} \cdot P(\omega_N) + \lambda_{NP} \cdot P(\omega_P)$

$$\text{Risk} = TPR \cdot P(\omega_P)\,(\lambda_{PP} - \lambda_{NP}) + FPR \cdot P(\omega_N)\,(\lambda_{PN} - \lambda_{NN}) + \lambda_{NN} \cdot P(\omega_N) + \lambda_{NP} \cdot P(\omega_P) \quad \rightarrow (4.30a)$$

The above equation has two terms that vary with TPR and FPR respectively, and one term that does not depend on either $(\lambda_{NN} \cdot P(\omega_N) + \lambda_{NP} \cdot P(\omega_P))$

Looking at the coefficient of TPR:

$$P(\omega_1)(\lambda_{PP} - \lambda_{NP})$$

Intuitively:

λ_{PP} should be zero - there is no cost to correctly predicting an edible mushroom.

λ_{NP} should be small or zero - there is minimal risk in incorrectly identifying an edible mushroom as inedible.

$$\therefore \boxed{\lambda_{PP} - \lambda_{NP} \leq 0} \rightarrow (4.3b)$$

Looking at the coefficient of FPR:

$$P(\omega_N) \cdot (\lambda_{PN} - \lambda_{NN})$$

Intuitively

λ_{PN} must be large - the risk of incorrectly classifying a poisonous mushroom is very high

λ_{NN} should be zero - there is no risk to correctly identifying a poisonous mushroom.

$$\therefore \boxed{\lambda_{PN} - \lambda_{NN} > 0} \rightarrow (4.3c)$$

From (4.3b) and (4.3c), we see that the risk increases with increasing FPR, but decreases with increasing TPR.

(4) We could inspect the dataset and calculate the following metrics by counting the samples within the dataset:

$N \rightarrow$ Total number of samples

$N_P \rightarrow$ Number of edible samples (non-poisonous)

$N_N \rightarrow$ Number of inedible samples (poisonous)

TP $P_{PP} \rightarrow$ Number of edible samples correctly classified

FN $P_{NP} \rightarrow$ Number of edible samples incorrectly classified

TN $P_{NN} \rightarrow$ Number of inedible samples correctly classified

FP $P_{PN} \rightarrow$ Number of inedible samples incorrectly classified

Definitions:

$\omega_P \rightarrow$ Class of edible mushrooms (non-poisonous)

$\omega_N \rightarrow$ Class of inedible mushrooms (poisonous)

Given the above metrics and definitions, we can calculate prior probabilities, TPR and FPR as shown below:

Prior Probabilities:

$$P(\omega_P) = \frac{\text{Number of edible mushrooms}}{\text{Total number of samples}} = \frac{N_P}{N} \quad \rightarrow (4.4a)$$

$$P(\omega_N) = \frac{\text{Number of inedible mushrooms}}{\text{Total number of samples}} = \frac{N_N}{N} \quad \rightarrow (4.4b)$$

$$\text{True Positive Rate} = \frac{\text{Number of correct edible predictions}}{\text{Number of actual edible mushrooms}}$$

$$= \frac{P_{PP}}{N_P}$$

We also know that every edible mushroom has P_{PP} correct classifications an P_{NP} incorrect classifications

$$\therefore \quad TPR = \frac{P_{PP}}{N_P} = \frac{P_{PP}}{P_{PP} + P_{NP}} \quad \rightarrow (4.4c)$$

$\frac{TP}{TP+FN}$

Similarly,

$$\text{False Positive Rate} = \frac{\text{Number of incorrect edible predictions}}{\text{Number of actual inedible mushrooms}}$$

$$FPR = \frac{P_{PN}}{N_N} = \frac{P_{PN}}{P_{NN} + P_{PN}} \quad \rightarrow (4.4d)$$

$\frac{FP}{TN+FP}$

(5)

Using (4.4 a–d) in (4.3a), we get:

$$\text{Risk} = \frac{P_{PP}}{N_P} \cdot \frac{N_P}{N} (\lambda_{PP} - \lambda_{NP}) +$$

$$\frac{P_{NP}}{N_N} \cdot \frac{N_N}{N} (\lambda_{PN} - \lambda_{NN}) +$$

$$\lambda_{NN} \cdot \frac{N_N}{N} + \lambda_{NP} \cdot \frac{N_P}{N}$$

$$\text{Risk} = \frac{1}{N} \cdot \left[P_{PP} (\lambda_{PP} - \lambda_{NP}) + P_{PN} (\lambda_{PN} - \lambda_{NN}) + \lambda_{NN} \cdot N_N + \lambda_{NP} \cdot N_P \right]$$

$$= (1/N) \left[\lambda_{PP} \cdot P_{PP} - \lambda_{NP} \cdot P_{PP} + P_{PN} \cdot \lambda_{PN} - P_{PN} \cdot \lambda_{NN} + \lambda_{NN} (P_{NN} + P_{PN}) + \lambda_{NP} (P_{PP} + P_{NP}) \right]$$

$$= (1/N) \left[\lambda_{PP} \cdot P_{PP} + \lambda_{PN} \cdot P_{PN} + \lambda_{NN} \cdot P_{NN} + \cancel{\lambda_{NN} P_{PN}} + \cancel{\lambda_{NP} P_{PP}} + \lambda_{NP} P_{NP} - \cancel{\lambda_{NP} \cdot P_{PP}} - \cancel{\lambda_{NN} P_{PN}} \right]$$

$$\boxed{\text{Risk} = (1/N) \left[\lambda_{PP} P_{PP} + \lambda_{PN} \cdot P_{PN} + \lambda_{NN} \cdot P_{NN} + \lambda_{NP} P_{NP} \right]}$$

→ (4.5a)

(4.5a) Is the average risk of the classifier on the dataset.

(6)

We can see from the analysis from the answer to problem 4.3 that the overall risk decreases as TPR increases, and it increases as FPR increases. With this in mind, let's consider the following cases:

Compare A and C:

A and C have the same TPR, but C has a higher FPR, which means that C has a higher risk. Therefore A is better.

Compare B and C:

B and C have the same FPR, but B has a higher TPR, which means that it has a lower overall risk. Therefore B is better.

(7) If we use $\lambda_{PP} = \lambda_{NN} = 0$ and $P(\omega_p) = P(\omega_n) = 0.5$ in (4.3a), we get:

$$\text{Risk} = TPR \times 0.5 \times (0 - \lambda_{NP}) + FPR \times 0.5 \times \lambda_{PN} + \lambda_{NP} \times 0.5$$

$$= (0.5)\left[FPR \cdot \lambda_{PN} - TPR \cdot \lambda_{NP} + \lambda_{NP} \right]$$

$$\boxed{\text{Risk} = (0.5)\left[\lambda_{PN} \cdot FPR + \lambda_{NP}\,(1 - TPR) \right]}$$

(4.7a)

Intuitively, $\lambda_{PN} \gg \lambda_{NP}$. It follows then that any risk reduction obtained by decreasing FPR will be much larger than the reduction obtained by increasing TPR.

Given that A has a lower FPR than C while B has a higher TPR than C, A should be preferable because it decreases FPR.

Question 5

(a)

This code demonstrates two seemingly independent ideas that may be connected as we address the remaining parts of question (5).

Part 1:

It defines "true" distributions of a 2-dimensional feature vector, conditioned on two classes positive (Seabass) and negative (Salmon). These are defined as multivariate Gaussian distributions (rv_pos, rv_neg) having defined means (ex: [1,3]) and covariance matrices (ex: $\begin{bmatrix} 1 & 0.8 \\ 0.8 & 1 \end{bmatrix}$) respectively. These values are visualized using a scatter plot

Estimate feature vectors are then sampled ($n = 1000$) from these distributions using a prior probability of $p = 0.5$. The feature vectors and the corresponding class labels are stored in X and y respectively. The estimated mean and covariance are also computed.

Part 2:

An ROC curve is simulated using a naive classifier that disregards feature vectors and simply selects scalars with probability of q where $q \in \{0, 0.1, 0.2 \ldots\ldots 1\}$. The resulting TPR and FPR are calculated for each q and are used to generate an ROC curve that is a 45° line (TPR = FPR). This is also visualized using a line graph.

(b) Please see attached code.

(c) The separating surfaces for both classifiers are linear. The first classifier is perpendicular to the vector connecting the means, while the second is not perpendicular, but is skewed based on the covariance matrix.

(d) Please see attached ROC curves. There are two curves, differing only in the underlying implementation. The first uses a general

Gaussian distribution initialized using identity and shared covariance matrices.

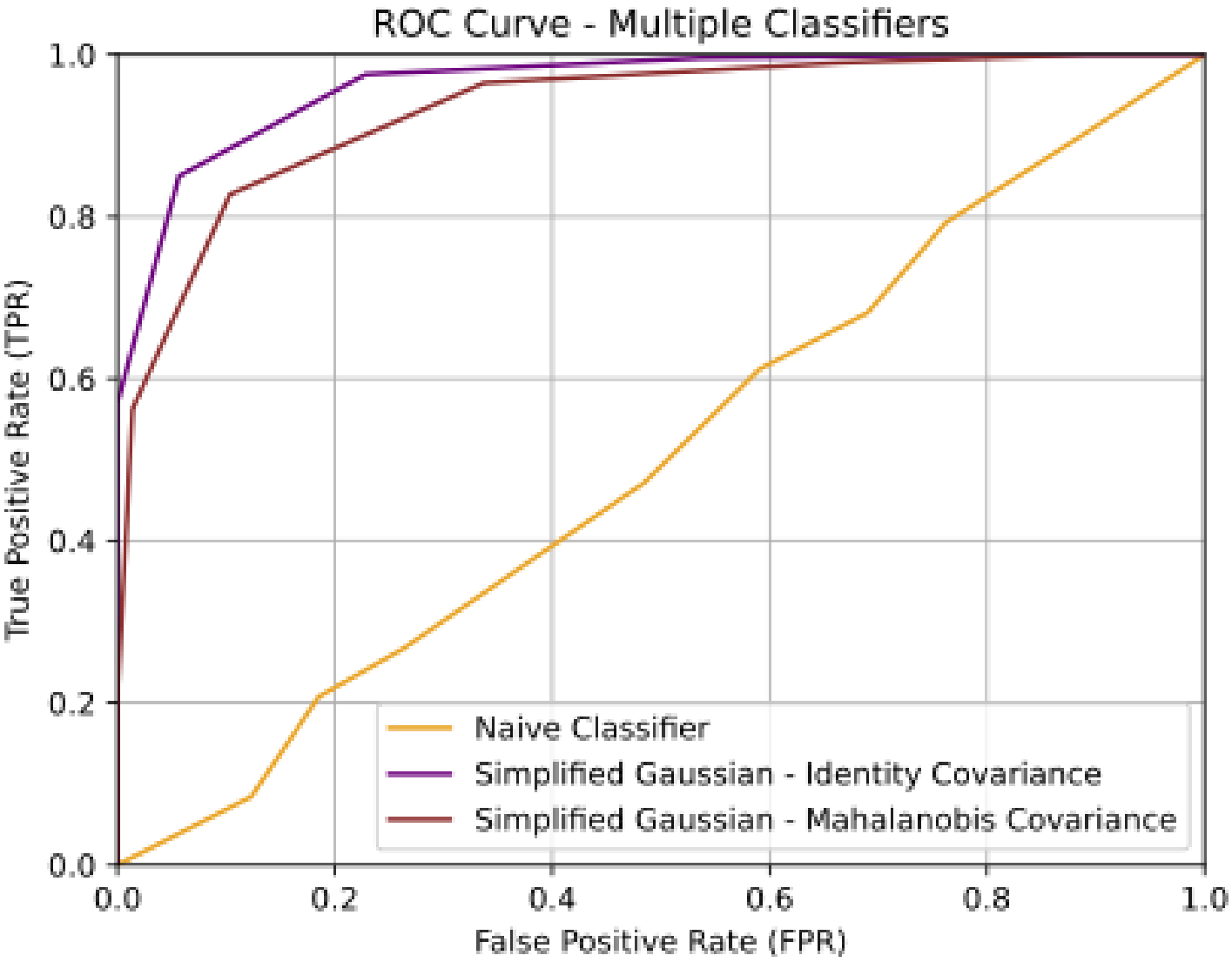
The second uses simplified expressions for the same two cases.

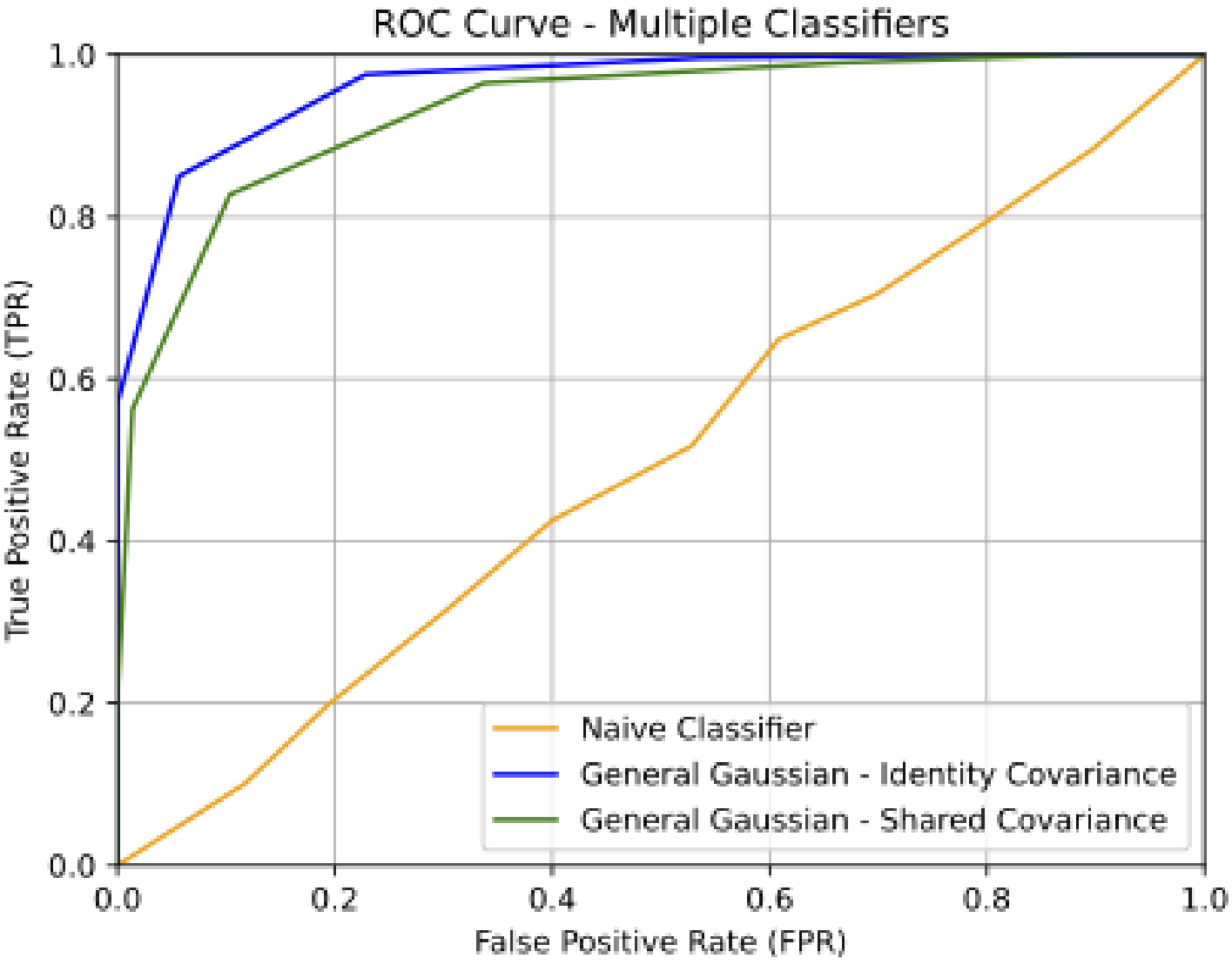
The resulting curves are identical.

Code: https://github.com/vamship/eece5644-homework/blob/master/hw1/src/final/question_5.py

Note on AI Usage:

- GenAI (Claude Sonnet 5.5) was used in analyzing the code from problem 5, and to verify my understanding of what the code was trying to do. I also used the LLM to confirm my understanding of the concepts by playing back to it assertions and asking it to confirm.
- When writing code, Github Copilot was used to auto-complete the code that I was writing. However, this was restricted to code completion only - I typed out the scaffold and the structure and only accepted CoPilot suggestions when it matched exactly what I was going to type out.


ROC Curve - Multiple Classifiers
True Positive Rate (TPR)
False Positive Rate (FPR)
0.0
0.2
0.4
0.6
0.8
1.0
Naive Classifier
Simplified Gaussian - Identity Covariance
Simplified Gaussian - Mahalanobis Covariance


ROC Curve - Multiple Classifiers
True Positive Rate (TPR)
False Positive Rate (FPR)
0.0
0.2
0.4
0.6
0.8
1.0
Naive Classifier
General Gaussian - Identity Covariance
General Gaussian - Shared Covariance